

Lecture-18
Mathematics 1 (15B11MA111)

CO [C105.4]

Module: Vector Differentiation

Topics to be covered

- **Gradient**

- **Geometric representation**

- **Examples**

- **Directional derivatives**

- **Example**

- **Maximum rate of change of scalar field**

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Gradient

Let $f(x, y, z)$ be a real valued function defining a scalar field. The gradient of this scalar field is defined as

$$\nabla f \text{ or } \text{grad } (f) = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$$

where $\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$ and read as **del** (operator).

Example 1: Find the gradient of the following scalar fields

(i) $f(x, y) = y^2 - 4xy$ at (1,2)

(ii) $f(x, y, z) = x^2y^2 + xy^2 - z^2$ at (3,1,1)

Solution: (i) $\nabla f = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} \right) f = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} \right) (y^2 - 4xy)$

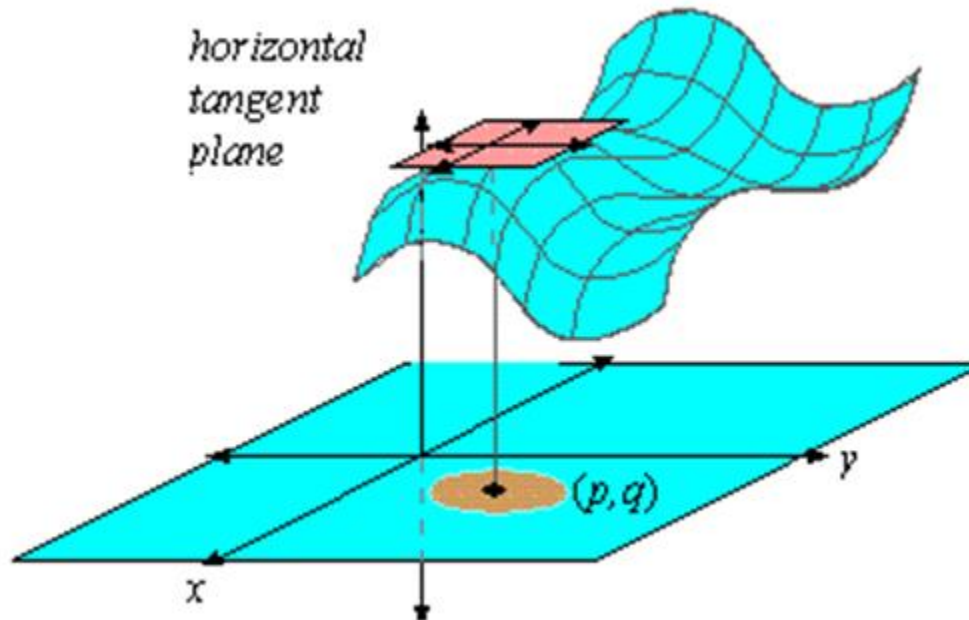
$$= -4y\hat{i} + (2y - 4x)\hat{j} = -8\hat{i} \text{ (at } 1, 2)$$

(ii) $\nabla f = (2xy^2 + y^2)\hat{i} + (2x^2y + 2xy)\hat{j} - 2\hat{k}$

$$\nabla f \text{ at } (3, 1, 1) = 7\hat{i} + 24\hat{j} - 2\hat{k}.$$

Geometric representation of the gradient

Let $f(x, y, z)$ be a differentiable scalar field and $f(x, y, z) = c$ be a level surface and $P_0(x_0, y_0, z_0)$ be a point on it. There are infinite number of smooth curves on the surface passing through P_0 and each of these curves has a tangent at P_0 . The totality of these tangent lines form a tangent plane to the surface at P_0 . A vector normal to this plane at P_0 is called the normal vector to the surface at this point.



Consider a curve C on the surface passing through a point P on the surface. Let $x = x(t), y = y(t), z = z(t)$ be a parametric representation of the curve C . Any point P on C has the position vector $\vec{r} = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$. Since the curve lies on the surface, we have

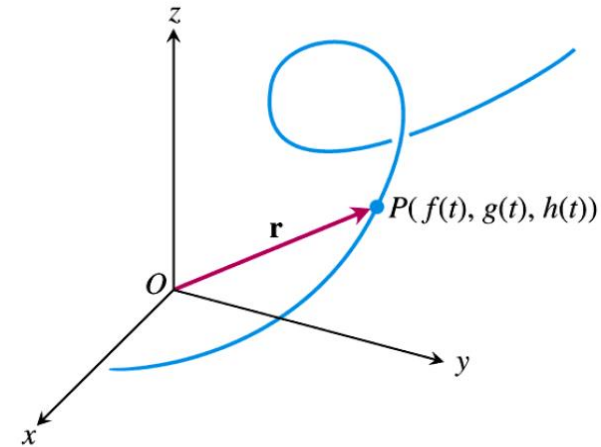
$$f(x(t), y(t), z(t)) = c.$$

Then, $\frac{d}{dt} f(x(t), y(t), z(t)) = 0$.

By chain rule, $\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} = 0$,

$$\left(\hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} \right) \cdot \left(\hat{i} \frac{dx}{dt} + \hat{j} \frac{dy}{dt} + \hat{k} \frac{dz}{dt} \right) = 0$$

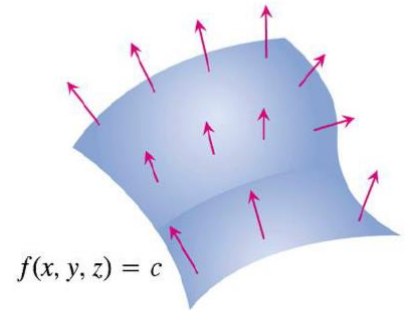
$$\nabla f \cdot \vec{r}'(t) = 0.$$



Let $\nabla f(P) \neq 0$ and $\bar{r}'(t) \neq 0$. Now $\bar{r}'(t)$ is a tangent vector to C at the point P and lies in the tangent plane to the surface at P . Hence, $\nabla f(P)$ is orthogonal to every tangent vector at P . Therefore, $\nabla f(P)$ is the vector normal to the surface $f(x, y, z) = c$ at the point P .

Remarks:

1. The unit normal vector is $\hat{n} = \frac{\text{grad} f}{|\text{grad} f|}$.
2. The gradient vector (normal vector) at a point $P_0(x_0, y_0, z_0)$ on the surface $f(x, y, z) = c$ can be used to derive the equation of tangent plane at P_0 , to the surface. Since the vector $(x - x_0)\hat{i} + (y - y_0)\hat{j} + (z - z_0)\hat{k}$ is a vector in the tangent plane, Hence it is orthogonal to the gradient vector at P_0 .



Therefore, $\nabla f(P_0) \cdot \{(x - x_0)\hat{i} + (y - y_0)\hat{j} + (z - z_0)\hat{k}\} = 0$

$$(x - x_0) \frac{\partial f}{\partial x}(P_0) + (y - y_0) \frac{\partial f}{\partial y}(P_0) + (z - z_0) \frac{\partial f}{\partial z}(P_0) = 0$$

is the equation of the tangent plane.

Example 2. Find a unit normal vector to the surface $xy^2 + 2yz = 8$ at the point $(3, -2, 1)$.

Solution: Let $f(x, y, z) = xy^2 + 2yz - 8$

$$\frac{\partial f}{\partial x} = y^2, \frac{\partial f}{\partial y} = 2xy + 2z \text{ and } \frac{\partial f}{\partial z} = 2y.$$

$$\text{Therefore, } \nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} = y^2 \hat{i} + 2(xy + z) \hat{j} + 2y \hat{k}$$

At $(3, -2, 1)$, $\nabla f = 4\hat{i} - 10\hat{j} - 4\hat{k}$. The unit normal vector at $(3, -2, 1)$ is

$$\frac{2\hat{i} - 5\hat{j} - 2\hat{k}}{\sqrt{33}}$$

Example 3. Find the normal vector and the equation of the tangent plane to the surface $z = \sqrt{x^2 + y^2}$ at the point (3,4,5).

Solution: Let $f(x, y, z) = z - \sqrt{x^2 + y^2} = 0$ be the surface. The normal is given as

$$\begin{aligned}\nabla f &= \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} = (\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z})(z - \sqrt{x^2 + y^2}) \\ &= -\frac{x}{\sqrt{x^2 + y^2}}\hat{i} - \frac{y}{\sqrt{x^2 + y^2}}\hat{j} + \hat{k} = -\frac{x}{z}\hat{i} - \frac{y}{z}\hat{j} + \hat{k}.\end{aligned}$$

At (3,4,5), the normal vector is

$$\nabla f(3,4,5) = -\frac{3}{5}\hat{i} - \frac{4}{5}\hat{j} + \hat{k}$$

and the tangent plane is given as

$$-\frac{3}{5}(x-3) - \frac{4}{5}(y-4) + (z-5) = 0, \text{ or } 3x + 4y - 5z = 0$$

Example 4: Find the angle between the surface $x \log z = y^2 - 1$ and $x^2 y = 2 - z$ at the point $(1,1,1)$.

$$\text{Ans: } \theta = \cos^{-1} \left(\frac{1}{\sqrt{30}} \right)$$

Directional Derivative

Let $f(P) = f(x, y, z)$ be a differentiable scalar field and $\hat{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$ be any unit vector. Then the directional derivative of f in the direction of $\hat{b}$ is given by

$$\text{directional derivative } (D_{\hat{b}}(f)) = \nabla f \cdot \hat{b} = \text{grad}(f) \cdot \hat{b}$$

Example 5: Find the directional derivative of $f(x, y) = x^2y^3 + xy$ at $(2, 1)$ in the direction of a unit vector which makes an angle of $\pi/3$ with the x - axis.

Solution: We have $\nabla f = (2xy^3 + y)\hat{i} + (3x^2y^2 + x)\hat{j}$

At $(2, 1)$, $\nabla f = 5\hat{i} + 14\hat{j}$.

The unit vector is given by $\hat{b} = \cos\theta\hat{i} + \sin\theta\hat{j} = \frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}$, for $\theta = 60$. Therefore the directional derivative is

$$D_b(f) = (5\hat{i} + 14\hat{j}) \cdot \left(\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right) = \frac{5+14\sqrt{3}}{2}.$$

Maximum rate of change of a scalar field

We have $D_b(f) = \nabla f \cdot \hat{b} = |\nabla f| \cdot |\hat{b}| \cos \theta = |\nabla f| \cos \theta$

where θ is the angle between the vectors ∇f and $\hat{b}$. Since $-1 \leq \cos \theta \leq 1$, *which implies that* $-|\nabla f| \leq D_b(f) \leq |\nabla f|$.

The maximum value of directional derivative is $|\nabla f|$ and it occurs when $\theta = 0$ that is $\hat{b}$ has direction of ∇f .

And the minimum value of directional derivative is $-|\nabla f|$ and it occurs at $\theta = \pi$, that is ∇f and $\hat{b}$ have opposite directions.

Conservative field

A vectors field $\vec{v}$ is called conservative if the vector function can be written as the gradient of a scalar function f , *i. e.* $\vec{v} = \nabla f$.

In such a field the work done in moving a particle from one place to another is independent of path.

Example 6. Show that the vector field defined by the vector function $\bar{v} = xyz(yz\hat{i} + xz\hat{j} + xy\hat{k})$ is conservative.

Solution: If the given vector field is conservative then it can be expressed as the gradient of a scalar function $f(x, y, z)$. Hence

$$\nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} = \bar{v} = xyz(yz\hat{i} + xz\hat{j} + xy\hat{k})$$

Comparing, we get

$$\frac{\partial f}{\partial x} = xy^2z^2, \frac{\partial f}{\partial y} = x^2yz^2, \frac{\partial f}{\partial z} = x^2y^2z.$$

Integrating the first equation, we get $f(x, y, z) = \frac{1}{2}x^2y^2z^2 + g(y, z)$.
Substituting in the second equation, we get

$$\frac{\partial f}{\partial y} = x^2 y z^2 = x^2 y z^2 + \frac{\partial g}{\partial y} \text{ or } \frac{\partial g}{\partial y} = 0 \text{ or } g = g(z).$$

Substituting in the third equation, we get

$$\frac{\partial f}{\partial z} = x^2 y^2 z = x^2 y^2 z + \frac{dg}{dz} \text{ or } \frac{dg}{dz} = 0 \text{ or } g = c, \text{ constant.}$$

Hence, $f(x, y, z) = \frac{1}{2} x^2 y^2 z^2 + c$. Therefore, there exist a scalar function $f(x, y, z)$ such that $\nabla f = \bar{v}$ and the vector field is conservative.

Assignment

- (i) Find the gradient of $\ln(x + y + z)$ at $(1, 2, -1)$
- (ii) Find the normal vector and the unit normal vector to the given surface (curve) $x^2 + y^2 = 25$ at $(3, 4)$.
- (iii) Find the directional derivative of $f(x, y, z) = x^2 + y^2 + 2z^2$ at $(1, 1, 2)$ in the direction of $\text{grad } f$.

Ans: (i) $(\hat{i} + \hat{j} + \hat{k})/2$ (ii) $2(3\hat{i} + 4\hat{j}), \frac{3\hat{i} + 4\hat{j}}{5}$ (iii) $2\sqrt{18}$

Thank you